

Homework (Habanero)

- Q1.** What is the equation of a line that passes through the point $(2, 3)$ with a slope of 4 in point-slope form?
- (A) $y - 3 = 4(x - 2)$
(B) $y = 4x - 5$
(C) $x - 2 = 4(y - 3)$
(D) $y = 3x + 1$
(E) $y = 4x - 1$
- Q2.** A patient's temperature T in degrees Fahrenheit is related to the time t in hours by the equation $T = 2t + 98.6$. What is the equation in standard form?
- (A) $2t + T = 98.6$
(B) $t - 0.5T = -49.3$
(C) $T - 2t = 98.6$
(D) $2t - T = -98.6$
(E) $-2t + T = 98.6$
- Q3.** What is the slope-intercept form of the line $x + 2y = 5$?
- (A) $y = -0.5x + 2.5$
(B) $y = 2x + 1$
(C) $y = -2x + 5$
(D) $y = 0.5x - 2.5$
(E) $y = x - 2.5$
- Q4.** A dosage of medicine is related to a patient's weight w in pounds by the equation $d = 0.2w + 5$. What does the slope represent in this context?
- (A) dosage increase per pound
(B) weight increase per dosage
(C) initial dosage
(D) weight decrease per dosage
(E) weight at which the dosage is zero
- Q5.** The equation of a line in slope-intercept form is $y = -3x + 2$. What is the equation in point-slope form if the line passes through $(0, 2)$?
- (A) $y - 2 = -3(x - 0)$
(B) $y = -3x + 2$
(C) $y + 2 = 3(x + 0)$
(D) $y - 0 = -3(x - 2)$
(E) $y + 3 = 2(x - 0)$
- Q6.** What is the y -intercept of the equation $y = 2x - 3$?
- (A) -3
(B) 2
(C) 3
(D) -2
(E) 1
- Q7.** The growth rate of a cell culture P is modeled by $P = 0.5t + 200$, where t is the time in hours. What is the growth rate?
- (A) 0.5 cells per hour
(B) 200 cells per hour
(C) $0.5 \cdot 200$ cells per hour
(D) t cells per hour
(E) $200t$ cells per hour
- Q8.** What is the equation of the line $y = 4$ in standard form?
- (A) $y = 4$
(B) $4y = 1$
(C) $0x - y = -4$
(D) $y - 4 = 0$
(E) $y + 0x = 4$

Open Ended Questions (FRQ)

- Q9.** The dosage d of a medicine is related to a patient's weight w by the equation $d = 0.1w + 2$. Graph this equation and label the y -intercept. If a patient's weight is 120 pounds, what dosage should be administered? Show all work and calculations.
- Q10.** The growth rate P of a cell culture is modeled by $P = 0.8t + 150$, where t is the time in hours. What is the growth rate? Use this information to determine the time at which the cell culture will have 500 cells. Show all work and calculations.

- Ensure students show all work for free response questions.
- Allow the use of graphing calculators for graphing.
- Remind students to label all axes and include units when graphing.